

JANSEVA-INFRA

Civic Issue Reporting and Management System

Project Documentation

Team Members

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1. Project Title

Janseva-Infra — Civic Issue Reporting and Management System

A comprehensive web application that enables citizens to report civic issues, track their status, and allows authorities to manage, prioritize, and resolve reported issues.

2. Team Members

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3. Problem Background

Civic issues such as damaged roads, potholes, garbage accumulation, broken streetlights, drainage problems, and other infrastructure-related concerns are often difficult for citizens to report and track effectively.

Traditional complaint systems can lack real-time tracking, transparency, proper prioritization, and efficient communication between citizens and authorities. Janseva-Infra provides a centralized digital platform where citizens can report problems with relevant details and images, while authorities can view, prioritize, assign, and manage these issues efficiently.

4. Problem Statement

There is a need for an efficient and transparent system that connects citizens with civic authorities for reporting and resolving local infrastructure issues.

- Lack of a centralized civic complaint platform.
- Difficulty in tracking the progress of complaints.
- Delayed identification of high-priority issues.
- Lack of visual and location-based information.
- Limited functionality when internet connectivity is unavailable.
- Difficulty for authorities in managing and assigning large numbers of complaints.

5. Impact

Janseva-Infra can create a more transparent, responsive, and efficient civic issue management process.

- **Citizens:** Easier reporting and real-time tracking of complaints.
- **Engineers:** Better visibility of assigned issues and their locations.
- **Supervisors:** Improved monitoring and management of civic complaints.
- **Authorities:** Faster identification and prioritization of important issues.
- **Community:** Improved accountability and potentially faster resolution of infrastructure problems.

The system also supports offline reporting, allowing issues to be stored locally and synchronized when connectivity becomes available.

6. Proposed Software Solution

We propose a web-based Civic Issue Reporting System with role-based access for **Citizens, Engineers, and Supervisors**.

Citizen Module: Register/login, report civic issues, upload photographs, provide issue details and location, and track reported issues.

Engineer Module: View assigned issues, access issue details and images, update progress/status, and use map-based information to locate issues.

Supervisor Module: Monitor reported issues, manage and assign issues, prioritize important complaints, and track overall issue resolution.

Smart Features: Gemini AI for issue-priority prediction, Leaflet maps, IndexedDB offline storage, automatic synchronization after reconnecting, and real-time issue status tracking.

7. Technical Feasibility

The project is technically feasible because it uses established web technologies and cloud services.

8. Operational Feasibility

The system is designed around the workflow of reporting, assigning, monitoring, and resolving civic issues.

Workflow: Citizen reports issue → Issue is stored → Priority is assessed → Authority/Engineer manages issue → Status is updated → Citizen tracks progress.

The application supports both online and offline workflows. When offline, an issue can be stored in an IndexedDB queue and synchronized with the backend once connectivity is restored.

9. Project Scope

In-Scope

- Citizen registration and authentication.
- Civic issue reporting and image attachment.
- Location-based issue visualization.
- Issue status tracking.
- Role-based access for Citizens, Engineers, and Supervisors.
- Assignment and management of issues.
- AI-based priority prediction.
- Offline issue reporting and automatic synchronization.

- Interactive maps and dashboards.

Out-of-Scope

- Direct integration with government municipal systems.
- Automated physical repair of civic infrastructure.
- IoT-based infrastructure monitoring.
- Large-scale deployment across multiple government departments.
- Advanced predictive infrastructure maintenance.
- Native Android/iOS applications.

10. Expected Outcome

The expected outcome of Janseva-Infra is a centralized civic issue management platform that improves communication between citizens and authorities.

1. Make civic issue reporting simpler and more accessible.
2. Improve transparency through issue status tracking.
3. Help authorities prioritize issues using AI-assisted assessment.
4. Improve issue management through role-based workflows.
5. Provide geographical context through interactive maps.
6. Allow reporting even without an active internet connection.
7. Automatically synchronize offline reports once connectivity is restored.
8. Improve accountability by maintaining issue information and progress.

Overall: Janseva-Infra aims to transform civic complaint management from a fragmented reporting process into a structured, trackable, and technology-driven workflow.

Technology Stack

Component	Technology
Frontend	React 18 + Vite
Authentication	Firebase Authentication
Database	Firebase Firestore
Image Storage	Supabase
Offline Storage	IndexedDB
Maps	Leaflet + React-Leaflet
Routing	React Router
State Management	React Context
HTTP Client	Axios
AI	Google Generative AI (Gemini)

Component	Technology
Charts	Recharts
Forms	React Hook Form